

TPU Simulator Parameter Sweep Configuration Reference

This document lists the detailed sweep configurations extracted from sweep.py. The simulation covers multiple groups, testing core parameter sensitivity under varying architectural, memory, and software workloads for both Llama 8B and 70B models. Every run isolates variations by overriding specific parameters against the default configuration (configs/default.yaml).

Section 1: Compute Sensitivity Groups

Group ID	Group Name	Parameter Variations Covered	Description / Methodology Notes
1a	1a_array_size	array_rows, array_cols: 64, 128, 256, 512 tile_rows, tile_cols matching size	Enforces tile size = array size to guarantee the tiler acts as a passthrough. Isolates pure array-size performance trends and hardware utilization scales without sub-tiling or padding artifact overheads.
1b	1b_systolic_units	systolic_units: 1, 2, 3, 4	Varies total parallel systolic processing units to evaluate compute scaling behavior and cross-unit execution imbalance overheads.
1c	1c_bidirectional	bidirectional: False, True	Compares performance impact of structural cell capabilities. bidirectional=True doubles effective capacity to 2 MACs/cell/cycle vs 1 MAC/cell/cycle standard unidir baseline.
1d	1d_vector_cores	vector_cores: 1, 2, 3, 6	Varies counts to trace non-systolic operational ceilings. Profiling baseline reveals vector core 0 sweeps ~37% of workloads (e.g., softmax/GeLU activation layers) while others share remainder, showing diminishing returns past 3 units.

Group ID	Group Name	Parameter Variations Covered	Description / Methodology Notes
1e	1e_simd_exp	Combos of (vec_simd, exp_lat): (16, 4), (32, 4), (64, 2), (64, 4), (64, 8), (128, 4)	Evaluates non-linear dependencies. Traces narrow/wide SIMD width alongside processing latencies for exponential arithmetic (e.g., softmax). Identifies attention-heavy context performance penalties.
1f	1f_access_bw	access_core.bandwidth: 32, 64, 128	Measures performance effects across scalar and non-contiguous memory activities (index lookup logic, gather steps). Validates low execution footprint (~4% utilization) empirically.

Section 2: Memory Subsystem Sensitivity Groups

Group ID	Group Name	Parameter Variations Covered	Description / Methodology Notes
2a	2a_hbm_bw	hbm_bw_tb_s: 0.5 (LPDDR5X), 1.0 (TPUv4), 2.0 (A100), 3.35 (H100)	Benchmarks peak memory-bandwidth bounds by replicating standard profiles across mobile memory standards up to server-class accelerators. Runs in both prefill and decode modes.
2b	2b_hbm_lat	hbm_lat_cycles: 50, 100, 200, 400	Sweeps underlying device access execution delays to stress latency tolerance and queue depth parameters within pipelines.
2c	2c_dma_channels	dma_channels: 1, 2, 4 <i>(Locks systolic_units=2)</i>	Fixed to 2 systolic processing units to force parallel workload profiles. Measures efficiency gains from expanded concurrency in memory transfer mechanisms.
2d	2d_stage_do	stage_double_buffer: :	Quantifies data prefetch interleaving. OFF runs serialized DMA-to-compute

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	ublebuf	False, True	steps (producing large roofline efficiency gaps); ON executes concurrent loading of tile N+1 during execution of tile N.
2e	2e_sram_ibuf	ibuf_kb: 512, 1024, 2048, 4096 (Locks model_sram=True)	Locks structural capacity enforcement limits. Forces internal DMA stalls when transient hardware footprint exceeds local cache capacity bounds to map on-chip SRAM budget pressure.

Section 3: Software & Workload Sensitivity Groups

Group ID	Group Name	Parameter Variations Covered	Description / Methodology Notes
3a	3a_prompt_len	prompt_len: 8B: 128, 512, 2048, 8192 70B: 128, 512, 2048	Maps execution transitions as computational split shifts from Feed-Forward Networks (FFN) to Attention structures. Tracks performance changes across context regimes (e.g., Llama-3-8B crossover threshold hits at ~21K tokens).
3b	3b_tile_size	tile_rows, tile_cols: 128, 256, 512 (Base array fixed at 256)	Isolates scheduler tiling impacts: Sub-tile (128) forces underutilized matrix packing; Exact fit (256) bypasses decomposition layers; Oversize (512) triggers internal iterative passes per instruction.
3c	3c_kv_cache	kv_cache_enabled: True, False	Measures total computational and memory overhead deltas caused by caching token states vs re-evaluating sequences.

Group ID	Group Name	Parameter Variations Covered	Description / Methodology Notes
3d	3d_head_dim	Combos of (head_dim, hidden_dim): (64, 2048), (128, 4096), (256, 8192) (Locks num_q_heads=32)	Modulates attention shape ratios while keeping projection structures constant. Larger head dimensions increase attention density, shifting the workload toward compute limits.
3e	3e_decode_ctx	prompt_len (context size): 512, 1024, 2048, 4096, 8192 (Locks mode="decode", gen_steps=32)	Decode-only evaluation loop. Directly captures how token generation throughput degrades under expanding KV cache memory footprints (calculates active allocation scales up to 8.39GB).

Section 4: Structural Architecture & Strategy Groups

Group ID	Group Name	Parameter Variations Covered	Description / Methodology Notes
4a	4a_gqa_8b	num_kv_heads, gqa_group_size: 32 (MHA, group=1), 8 (GQA-4, group=4), 4 (GQA-8, group=8), 2 (GQA-16, group=16), 1 (MQA, group=32)	Sweeps Grouped-Query Attention strategies over the Llama 8B configuration in both prefill and decode regimes. Charts the memory bandwidth savings gained when reducing active KV footprints down to Multi-Query boundaries. Explores 32 generation steps during decode tasks.
4b	4b_gqa_70b	num_kv_heads, gqa_group_size: 64 (MHA, group=1), 8 (GQA-8, group=8), 4 (GQA-16, group=16), 1 (MQA,	Applies equivalent GQA variations tailored for the larger 70B parameter configuration model bounds. Runs in both prefill and decode modes.

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		group=64)	
5	5_pareto	3×3 grid combining: array_rows/cols (128, 256, 512) × hbm_bw_tb_s (1.0, 2.0, 3.35)	A 9-cell grid mapping compute limits vs memory bottlenecks. Locks tile size to array size across all variations to bypass confounding subtiling edge behaviors, tracking the exact inflection ridge where bound states flip.
6	6_calibration	Architectural targets for: H100_SXM5, A100_SXM4, TPUv4_chip Crossed with stage_double_buffer (False, True)	Validation baseline group. Bypasses absolute TPS comparisons to target hbm_util_pct directly against published chip sheets. Models real-world hardware decode properties (50-70% bandwidth utilization limits at batch size 1) to verify simulator accuracy within ±20%.